

Project Overview

This project explored the relationship between climate anomalies and wildfire activity in the western United States from 2000–2023. Using historical wildfire and climate datasets, I analyzed whether rising temperatures and changing precipitation patterns were associated with increases in wildfire frequency across California, Arizona, and Oregon.

The project combined environmental data analysis, statistical visualization, and computational methods to investigate how climate variability may influence wildfire behavior over time.

Methods

Wildfire data was obtained from the National Interagency Fire Center, while climate anomaly data was sourced from NOAA. Temperature and precipitation anomalies were averaged annually for each state and merged with yearly wildfire counts.

Using Python analysis tools, I:

- Cleaned and organized multi-source environmental datasets
- Visualized long-term wildfire and climate trends
- Conducted correlation analysis between wildfire frequency and climate anomalies
- Created boxplots and heatmaps to compare climate variability across states
- Interpreted ecological and environmental implications of observed trends

Results

The analysis showed that wildfire counts did not consistently increase alongside rising temperatures. While temperature anomalies generally increased over time and precipitation became more variable, the statistical relationships between climate variables and wildfire frequency were relatively weak.

California displayed the greatest climate variability among the states analyzed, suggesting that climate extremes may contribute more strongly to wildfire severity than wildfire frequency alone.

The project highlighted that wildfire behavior is influenced by multiple interacting factors beyond climate, including land management practices, human development, and fire suppression strategies.

Conclusion

This project demonstrated the complexity of environmental systems and the challenges of interpreting climate-driven ecological change using real-world datasets. It also emphasized the importance of selecting meaningful metrics when studying wildfire impacts, suggesting that future analyses involving acres burned, fire intensity, or predictive modeling may provide stronger insights into climate–wildfire relationships.